

ASHISH AGRAWAL

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Systems Engineer transitioning from over 15 years of global operational leadership, to low-level systems programming. Focused on the layer where hardware-aware, low-latency C++ and Linux systems programming meet applied AI and NLP coursework. Bringing over a decade of high-stakes problem-solving to concurrent, real-time, and performance-critical software.

TECHNICAL SKILLS

- **Languages:** C++, C, Python, Java, Bash, x86 Assembly
- **Systems:** Multithreading, Concurrency, Real-Time Systems (SCHED_FIFO), IPC, Memory Management
- **AI/NLP Coursework:** Machine Learning, N-gram Language Models, LSTM Encoder-Decoder with Attention, Transformer Fine-Tuning (DistilGPT-2), RAG Pipelines (LangChain, FAISS)
- **Software Design:** Object-Oriented Programming, Design Patterns (Composite, Builder, Proxy, Observer)
- **Tools:** Linux, Git, Docker, Make/CMake, Vim

PROJECTS

Safety-Critical Multithreaded Watchdog | C++

- Built a real-time monitoring system designed to trigger an immediate Emergency Stop (E-Stop) if the primary control software hangs or misses a strict 500ms heartbeat.
- Prevented deadlocks and priority inversion by using a lock-free communication path with C++ atomics, ensuring the monitor thread stays independent of the main process.
- Stress-tested for deterministic failure by developing a fault-injection suite that simulated CPU starvation and thread crashes to guarantee the system fails safely under pressure.

Tech: C++20, POSIX Threads (pthread), Atomics, Linux (SCHED_FIFO)

Lock-Free Concurrent Queue | C

- Developed a high-throughput, non-blocking queue using the Michael & Scott algorithm to remove performance bottlenecks typically caused by mutex contention.
- Ensured thread-safety across 32+ concurrent threads by implementing atomic primitives and compare_exchange_strong for linearizable operations.
- Solved complex memory management issues by implementing a deferred reclamation strategy, validating the entire system through a suite of 190+ automated test cases.

Tech: C, Atomics, Memory Barriers, Concurrency Testing

Multi-Type Financial Fraud Detector | Python

- Created a fraud detection system that automatically recognizes data types to spot credit card fraud, tax evasion, and invoice errors across financial sectors.
- Used an Isolation Forest model (unsupervised anomaly detection) and custom rules to flag suspicious patterns and assign a 0–100 risk score, helping auditors focus on the most critical threats first.
- Developed an interactive dashboard using Streamlit and Plotly to show real-time fraud trends and export reports for compliance and investigation.

Tech: Python, Scikit-learn, Streamlit, Pandas, Plotly, NumPy

PROFESSIONAL EXPERIENCE

Senior Program Manager | Amazon Air

Ontario, Canada | Aug 2022 – Sept 2023

- **ML & Automation:** Directed the implementation of internal ML-based automation models to optimize volume capacity planning across North American sites by partnering with Data Scientists and Software Engineers to train predictive algorithms on historical load data.

- Data Engineering: Engineered capacity analysis frameworks to resolve critical throughput bottlenecks by utilizing SQL queries to deep-dive massive Amazon Air datasets and collaborating with BIEs to visualize utilization gaps.
- Overhauled operational planning protocols to accelerate adoption of new research capabilities by executing internal "Go-to-Market" strategies and change management frameworks for tool deployment.

Site Operations Manager | Amazon Logistics

Ontario, Canada | Aug 2020 – July 2022

- Managed high-volume operations for a 500,000 sq. ft. facility, enhancing productivity by 22% and delivering over \$2.5M in projected cost savings.
- Architected network-level logistics programs to optimize load planning and resource allocation, achieving region-leading throughput of 51.2 TPH (throughput per hour).
- Led cross-functional teams in risk assessments to maintain system stability during peak high-volume periods.

Manager of Operations | Various Companies

Canada | Jul 2008 – Jul 2020

- Includes roles at Fresh Start Foods, Maxxam Analytics, and Vikram Laboratories.
- Handled operations across manufacturing and lab environments with a focus on process efficiency and fault tolerance.
- Refactored scheduling logic to eliminate bottlenecks, increasing output by 77% and uptime by 90%.
- Enforced rigorous QC standards to maintain zero data integrity failures in high-volume workflows and re-engineered procedures to cut setup times by 20%.

EDUCATION

Bachelor of Science (Honours), Computer Science

Trent University, ON | Expected April 2027

- Distinction: Dean's Honour Roll (2023–2024, 2024–2025)
- **Relevant Coursework:** Operating Systems, Systems Programming, Computer Organization, Computer Networks, High-Performance Computing (in progress), Advanced Software Architecture & Design, Applied AI & Machine Learning, LLMs & Natural Language Processing, AI in Information Systems, Data Mining, Discrete Structures, Linear Algebra, Calculus I/II, Probability I

Bachelor of Technology, Biotechnology

Rai Foundation Engineering College, India